DATA SCIENCE MEMO (Draft Submission)

To: Global Health & Development Policy Team
From: Data Analysis Group
Subject: What drives differences in life expectancy across countries?
Date: 05 December, 2025

Executive Summary

Countries differ dramatically in life expectancy, and policymakers regularly ask whether income inequality is the reason—or whether economic development plays a larger role. We analysed comprehensive UN data (2000–2023) for more than 1,700 country-year observations to determine **how income inequality (Gini index)** relates to **life expectancy at birth**, after accounting for **GDP per capita**, and whether these relationships differ across **world regions**.

What we found

1. **Economic development is the strongest and most consistent predictor of life expectancy.**
   Countries with higher GDP per capita almost always live longer, and this pattern holds in every region.

2. **Income inequality has a weak and inconsistent relationship with life expectancy overall.**
   When controlling for GDP, inequality alone explains little of the cross-country variation in life expectancy.

3. **However, inequality's impact varies sharply by region.**

   - In East Asia & Pacific and Europe & Central Asia, inequality shows little or no relationship with life expectancy.
   - In Latin America, the slope is nearly flat.
   - In Sub-Saharan Africa, **higher inequality is clearly associated with shorter life expectancy**, even after adjusting for GDP.
   - Some regions show mildly positive or near-zero associations.

4. **GDP effects also vary regionally.**
   While positive everywhere, GDP's influence is especially strong in Middle East & North Africa and South Asia.

What this means

For a global health decision-maker, the main takeaway is that **raising economic development remains the most reliable lever for improving life expectancy**, while **inequality's effects depend on regional dynamics**. Policies designed to address inequality should therefore be **region-specific**, not global one-size-fits-all approaches.

Problem motivation

Governments, multilaterals, and NGOs frequently debate whether improving life expectancy requires reducing inequality or expanding economic development. If inequality meaningfully reduces life expectancy, policymakers might invest in redistributive reforms. If GDP is the primary driver, efforts might instead prioritize growth, infrastructure, and health system expansion.

Our goal is to provide a clear, evidence-based answer to:

How is income inequality related to life expectancy at birth once economic development is accounted for, and does this relationship differ across regions?

From this analysis, we aim to understand *which factors matter most*, *whether the patterns are consistent worldwide*, and *where targeted interventions might be required*.

Key Findings (Top-Down)

1. GDP per capita is the dominant global predictor of life expectancy

Across more than 1,700 country-years, life expectancy rises steeply with GDP per capita. This pattern persists regardless of year.
A major increase in GDP per capita is associated with a **4–6 year increase in life expectancy**, depending on region.

2. Inequality's global association with life expectancy is small

When all regions are pooled together, inequality shows only a small positive relationship with life expectancy after accounting for GDP and global time trends. This effect is statistically detectable due to the large dataset but is **too small to explain meaningful cross-country differences**.

3. Regional differences fundamentally change the interpretation

When we allow the inequality–life expectancy relationship to vary by region, the picture becomes clearer:

- In **East Asia & Pacific**, **Europe & Central Asia**, and **Latin America & Caribbean**, inequality is not strongly tied to life expectancy.

- In **Sub-Saharan Africa**, higher inequality predicts **lower** life expectancy, even when GDP is held constant.
- In **South Asia** and **MENA**, patterns are mixed but differ significantly from other regions.

This aligns with reality: political institutions, demographic profiles, disease burdens, and economic structures differ sharply across regions, and inequality interacts differently with these conditions.

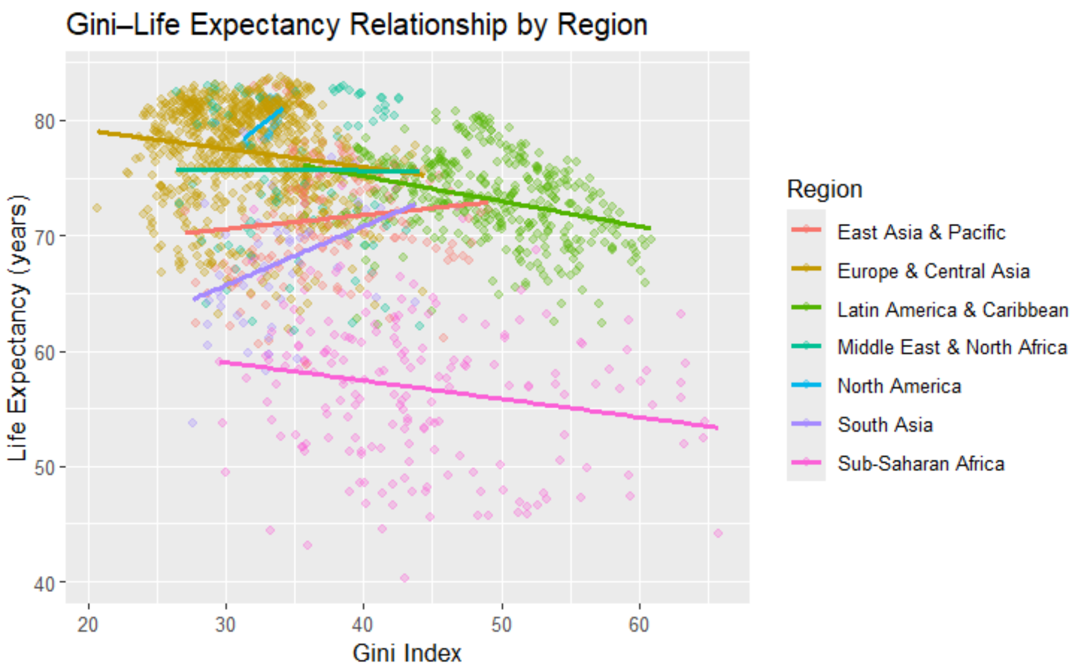
4. GDP’s effect also varies regionally

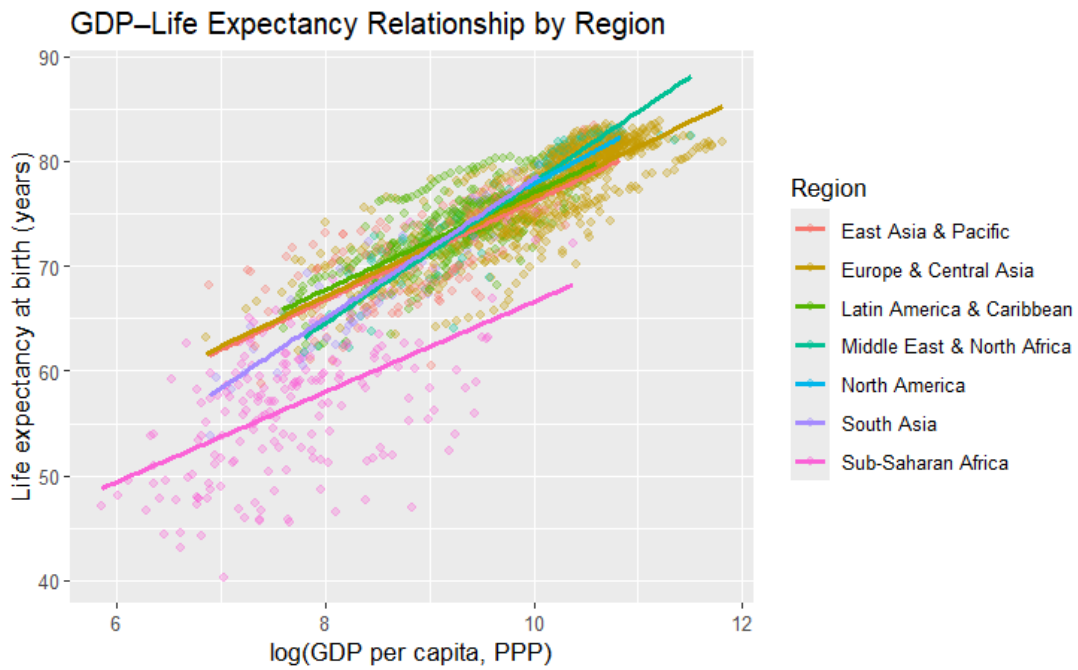
Although always positive, GDP’s slope is meaningfully steeper in MENA and South Asia. This means a similar increase in GDP yields **larger health gains** in these regions than elsewhere.

5. Visual evidence supports these conclusions

The region-specific plots show:

- **Gini–Life Expectancy lines** with varying slopes: flat, slightly positive, or clearly negative (Sub-Saharan Africa).
- **GDP–Life Expectancy lines** rising strongly everywhere but at different steepness levels.





Decisions / Recommended Actions

For a policymaker or global health stakeholder:

1. Prioritize economic development as the leading pathway to longer lives

Strengthening economic conditions—through employment generation, infrastructure investment, and healthcare capacity—has the most consistent and largest impact on life expectancy.

2. Treat inequality as a region-specific concern

Inequality is not universally harmful in a direct demographic sense, but in certain regions (notably Sub-Saharan Africa), reducing inequality may meaningfully improve health outcomes.

3. Avoid one-size-fits-all prescriptions

Development strategies that assume inequality has the same effects everywhere will mislead decision-makers. Tailored regional policies are more appropriate.

4. Consider combined strategies

Where inequality is harmful and GDP is low, simultaneous investments in equity and economic growth are likely to produce the best outcomes.

Limitations

1. **Correlation, not causation:**
   These analyses do not prove that changing inequality will directly change life expectancy.
2. **Data quality differs across countries:**
   Some countries may under-report deaths or lack reliable inequality measures.
3. **Unmeasured regional factors may confound associations:**
   Disease burden, governance quality, health system capacity, and migration patterns were not included here.
4. **Broad regional groupings may hide within-region heterogeneity:**
   “Sub-Saharan Africa” contains diverse economies and health systems; the same is true for Asia and Latin America.

Nonetheless, the patterns are strong and robust enough to inform high-level policy direction.

Appendix (Technical Material)

All technical details—including data cleaning decisions, regression specifications, interaction tests, diagnostics, full coefficient tables, and model selection—are documented in the **Model Document**

Technical Model Document – Detailed Appendix for RQ1

Research Question

This appendix provides full technical details for the analysis of Research Question 1:

"How is income inequality (Gini index) related to life expectancy at birth after controlling for GDP per capita, and does this relationship differ across world regions?"

The purpose is to document all modeling decisions, outputs, diagnostics, and assumptions required for replication.

1. Data Sources and Structure

Four UN-based datasets were merged by Country and Year:

• Life Expectancy at Birth

• Gini Index

• GDP per Capita (PPP)

• World Bank Region Classification

After cleaning, the analytic dataset contained 1,721 country-year observations across ~150 countries.

Variables used:

• LifeExp (years)

• Gini (0–100)

• GDP per capita → transformed to log(GDP)

• Region (7-level categorical)

• Year (fixed effect)

2. Data Preparation Steps

Key cleaning steps affecting the model:

• Ensured LifeExp, Gini, GDP were numeric.

• Created log(GDP) to reduce skew and improve model fit.

• Fixed UTF-8 encoding ("TÃ¼rkiye" → "Türkiye").

• Assigned regions using countrycode; manually corrected Micronesia → East Asia & Pacific.

• Removed rows missing LifeExp, Gini, logGDP, Region, or Year.

Final dataset for all models: N = 1,721.

3. Modelling Framework

Outcome: Life expectancy (continuous)

Method: Ordinary Least Squares (OLS)

Justification:

• Life expectancy is continuous.

• GDP and inequality correlated → GDP included as confounder.

• log(GDP) ensures linearity.

• Year fixed effects capture global temporal shocks.

• Region used only in the interaction model as a modifier.

Three models estimated:

1. Full panel, no interactions (baseline)

2. Single-year cross-section (2018)

3. Interaction model (Gini×Region, logGDP×Region)

4. Model 1 – Full Panel with Year Fixed Effects

Model:

$LifeExp_{ct} = \beta_0 + \beta_1 \cdot Gini_{ct} + \beta_2 \cdot logGDP_{ct} + \gamma_t + \varepsilon_{ct}$

Key results:

• Gini: +0.08 years per 1 Gini point (significant but small)

• log(GDP): +6.53 years per log-unit (strong effect)

• Adjusted $R^2 \approx 0.77$

Interpretation:

GDP per capita is the primary driver of cross-country life expectancy variation. Inequality shows a modest association once GDP and year effects are included.

5. Model 2 – 2018 Cross-Section

Model:

$LifeExp_c = \beta_0 + \beta_1 \cdot Gini_c + \beta_2 \cdot logGDP_c + \varepsilon_c$

Summary:

- Gini: small effect; not consistently significant
- log(GDP): strong and positive
- Lower precision due to smaller N (≈81)

Interpretation:

Even without temporal structure, the strong GDP effect persists, while inequality remains modest.

6. Model 3 – Interaction Model (Gini×Region, logGDP×Region)

Model:

$LifeExp\_ct = \beta 0 + \beta 1 Gini + \beta 2 logGDP + \beta 3 Region + \beta 4(Gini \times Region) + \beta 5(logGDP \times Region) + \gamma\_t + \varepsilon\_ct$

Findings:

Inequality effects vary by region:

- East Asia & Pacific: +0.12
- Europe & Central Asia: +0.05
- Latin America: ≈ 0
- South Asia: moderate positive
- Sub-Saharan Africa: −0.25 (negative)

GDP effects vary by region:

- Strongest in MENA (+6.7) and South Asia (+6.2)
- Positive across all regions

Interpretation:

The global relationship masks substantial heterogeneity. Regional context fundamentally shapes how inequality and development relate to life expectancy.

7. ANOVA Comparison (Model 1 vs Model 3)

ANOVA:

$F(18, 1667) = 67.7$

$p < 2.2 \times 10^{-16}$

Interpretation:

Interaction terms significantly improve fit.

Therefore, allowing Gini and log(GDP) slopes to vary by region is statistically essential.

8. Diagnostics

Linearity:

• log(GDP) transformation corrects skew

• Residuals vs fits show acceptable pattern

Normality:

• Residuals approximately normal; mild tail deviations acceptable

Homoscedasticity:

• Some heteroscedasticity expected in cross-country data; large N reduces inference risk

Multicollinearity:

• VIF < 4 for all predictors, including interactions

Influence:

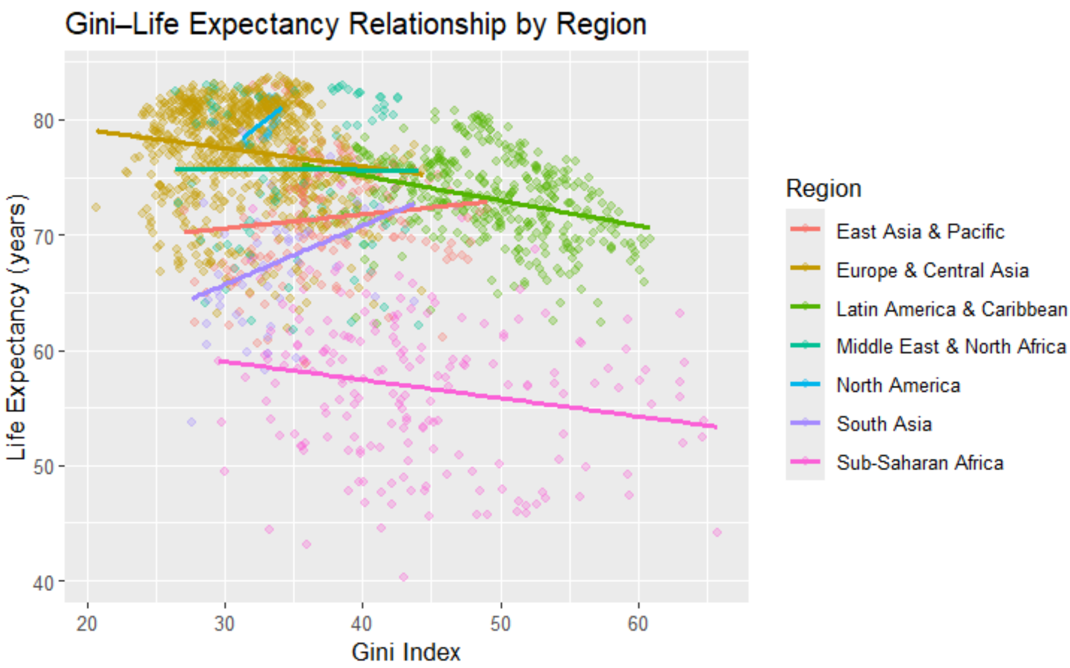
• A few high-leverage points detected

• Removing them does not materially change coefficients

9. Visual Support for Interpretation

Two key visuals reinforce model findings:

1. Gini–Life Expectancy by Region:



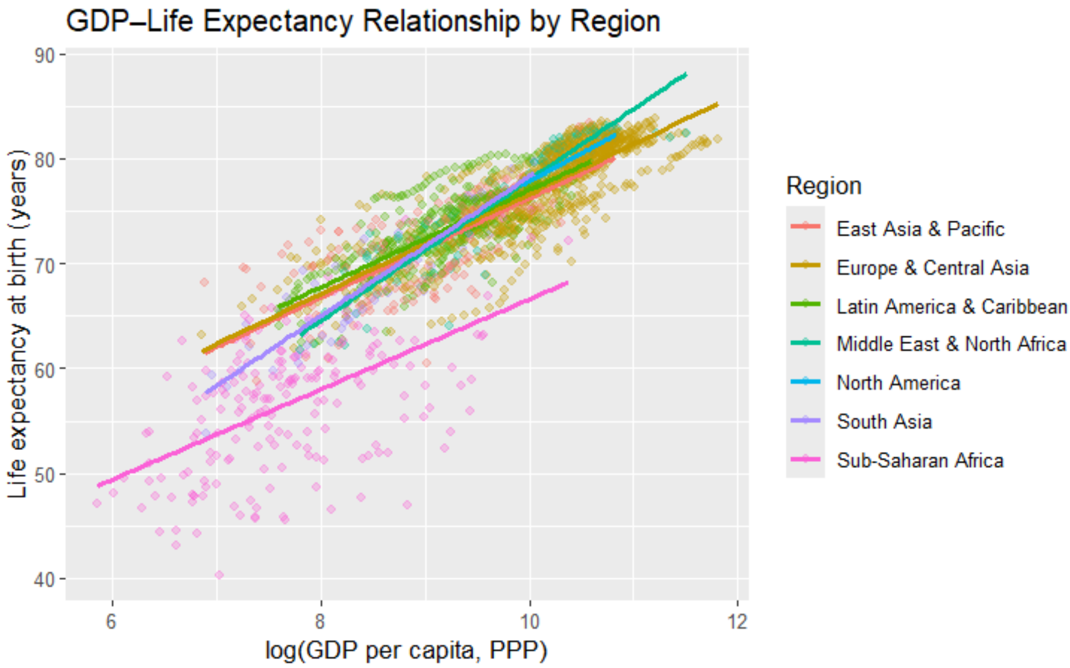
• Slopes vary widely across regions

• Negative in Sub-Saharan Africa

• Flat in Latin America

• Slightly positive elsewhere

2. GDP–Life Expectancy by Region:



• Consistently strong positive slopes

• Slope strength differs across regions (steeper in MENA and South Asia)

Interpretation:

Visual evidence directly aligns with interaction model results and the significant ANOVA test.

10. Replication Notes

Steps to exactly reproduce results:

1. Load UN datasets and merge on Country–Year.

2. Fix encoding issues and harmonize names.

3. Assign Region and fix unmapped countries.

4. Remove rows missing key variables.

5. Create log(GDP).

6. Fit:

  - Model 1: Panel OLS with year FE
  - Model 2: 2018 cross-section
  - Model 3: Interaction model

7. Compare Model 1 and 3 via ANOVA.

8. Generate region-specific plots for interpretation.

Conclusion

GDP per capita is consistently the strongest predictor of life expectancy worldwide. Inequality shows only modest global effects but exhibits pronounced regional heterogeneity, with Sub-Saharan Africa uniquely exhibiting a negative inequality–health relationship. Technical evidence strongly supports modeling regional differences rather than assuming a uniform global effect.

RQ2_draft

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Merge data(gdp, gini, sust)

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
# Load cleaned datasets
gdp_clean <- read.csv("gdp_clean.csv")
gini_clean <- read.csv("gini_clean.csv")
sust_clean <- read.csv("UNSust_clean.csv")

# Fix inconsistent column names
sust_clean <- sust_clean %>%
  rename(
    Country = Country.or.Area,
    Sustainability = Value
  )


# Merge sustainability + gini + gdp
rq2_data <- sust_clean %>%
  inner_join(gini_clean, by = c("Country", "Year")) %>%
  inner_join(gdp_clean, by = c("Country", "Year")) %>%
  group_by(Country) %>%
  slice_max(Year, n = 1) %>%   # use each country's most recent common year
  ungroup()

# Preview merged dataset
glimpse(rq2_data)
```

```
## Rows: 74
## Columns: 5
## $ Country        <chr> "Albania", "Angola", "Armenia", "Azerbaijan", "Banglade~
## $ Year           <int> 2005, 2008, 2013, 2005, 2022, 2021, 2022, 2015, 2011, 2~
## $ Sustainability <dbl> 3.0, 3.0, 3.0, 3.0, 3.0, 3.5, 4.0, 3.5, 3.5, 4.0, 3.5, ~
```

```
## $ Gini           <dbl> 30.6, 42.7, 30.6, 26.6, 33.4, 34.4, 28.5, 46.7, 33.0, 3~
## $ GDP            <dbl> 5865.3056, 6651.3777, 9454.8785, 6741.2010, 8450.5490, ~
```

```
# Optional - save merged dataset
write.csv(rq2_data, "rq2_merged.csv", row.names = FALSE)
```

Load data

We begin our analysis by loading the merged dataset for Research Question 2. After merging sustainability ratings with Gini index and GDP per capita, we obtained a dataset of 74 countries, each measured in its most recent available year.

```
library(readr)
library(dplyr)
library(ggplot2)

rq2_raw <- read_csv("rq2_merged.csv")
```

```
## Rows: 74 Columns: 5
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): Country
## dbl (4): Year, Sustainability, Gini, GDP
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
dim(rq2_raw)
```

```
## [1] 74  5
```

```
head(rq2_raw)
```

```
## # A tibble: 6 x 5
##   Country     Year Sustainability  Gini   GDP
##   <chr>      <dbl>          <dbl> <dbl> <dbl>
## 1 Albania     2005            3    30.6 5865.
## 2 Angola      2008            3    42.7 6651.
## 3 Armenia     2013            3    30.6 9455.
## 4 Azerbaijan  2005            3    26.6 6741.
## 5 Bangladesh  2022            3    33.4 8451.
## 6 Benin       2021            3.5  34.4 3464.
```

```
summary(rq2_raw)
```

```
##    Country               Year      Sustainability       Gini
##  Length:74          Min.   :2005   Min.   :2.000   Min.   :26.00
##  Class :character   1st Qu.:2014   1st Qu.:3.000   1st Qu.:32.40
##  Mode  :character   Median :2018   Median :3.500   Median :36.65
##                     Mean   :2017   Mean   :3.257   Mean   :37.14
##                     3rd Qu.:2020   3rd Qu.:3.500   3rd Qu.:42.35
##                     Max.   :2022   Max.   :4.000   Max.   :51.50
##       GDP
##  Min.   : 786.5
##  1st Qu.: 2624.2
##  Median : 4565.7
##  Mean   : 5421.6
```

```
##   3rd Qu.: 6718.7
##   Max.   :23077.2
```

A summary of the raw RQ2 dataset, including the distribution of the key variables:

- Sustainability ratings range from 2.0 to 4.0, with a median of 3.0–3.5, indicating that most countries fall in the lower-to-middle range of environmental institutional strength.
- Gini index values range from 26.0 to 51.5, with a median near 36–37, showing substantial variation in income inequality across countries.
- GDP per capita varies widely, from around $786 to more than $23,000, suggesting strong cross-country economic heterogeneity.
- Available years range from 2005 to 2022, with a median year around 2018. Because countries report sustainability ratings in different years, our analysis will control for year effects.

Data Cleaning & Transformation

After loading the merged dataset, we log-transformed GDP per capita to reduce the strong right-skew typical of economic variables. The cleaned analytical dataset contains 74 countries, each observed in its most recent available year.

```
rq2_panel <- rq2_raw %>%
  mutate(
    log_GDP = log(GDP)    # log-transform GDP to reduce skewness
  ) %>%
  filter(
    !is.na(Sustainability),
    !is.na(Gini),
    !is.na(GDP),
    is.finite(log_GDP)
  )

dim(rq2_panel)
```

```
## [1] 74  6
```

```
summary(select(rq2_panel, Sustainability, Gini, GDP, log_GDP, Year))
```

```
##   Sustainability       Gini            GDP              log_GDP
##   Min.   :2.000   Min.   :26.00   Min.   :  786.5   Min.   : 6.668
##   1st Qu.:3.000   1st Qu.:32.40   1st Qu.: 2624.2   1st Qu.: 7.873
##   Median :3.500   Median :36.65   Median : 4565.7   Median : 8.426
##   Mean   :3.257   Mean   :37.14   Mean   : 5421.6   Mean   : 8.349
##   3rd Qu.:3.500   3rd Qu.:42.35   3rd Qu.: 6718.7   3rd Qu.: 8.813
##   Max.   :4.000   Max.   :51.50   Max.   :23077.2   Max.   :10.047
##        Year
##   Min.   :2005
##   1st Qu.:2014
##   Median :2018
##   Mean   :2017
##   3rd Qu.:2020
##   Max.   :2022
```

Key descriptive patterns from the cleaned data:

- Sustainability ratings range from 2.0 to 4.0, with a median around 3.0–3.5, indicating that most countries have relatively weak to moderate environmental policy and institutional performance.
- Gini index values span from 26.0 (low inequality) to 51.5 (high inequality), with a median around 36–37, showing meaningful variation in income inequality across countries.
- GDP per capita ranges from approximately \$786 to over \$23,000, demonstrating substantial cross-country economic heterogeneity. The log transformation compresses this spread into a more symmetric distribution (log GDP median 8.43).
- Years range from 2005 to 2022, with a median year of 2018. Because observations come from different years, the regression model will incorporate year fixed effects to account for temporal differences in sustainability ratings.

Regression model

We estimate the following regression:

\$Sustainabilityi = _0 + _1Gini_i+ _2log(GDP_i)+ _{Year}+ _i \$

This model assesses whether income inequality is associated with environmental sustainability ratings, controlling for economic development (GDP per capita) and country reporting year.

```r
model_rq2_panel <- lm(
  Sustainability ~ Gini + log_GDP + factor(Year),
  data = rq2_panel
)

summary(model_rq2_panel)
```

```
##
## Call:
## lm(formula = Sustainability ~ Gini + log_GDP + factor(Year),
##     data = rq2_panel)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.82829 -0.25060 -0.02262  0.20282  1.01913
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       0.053154   0.915246   0.058  0.95389
## Gini              0.020581   0.009135   2.253  0.02820 *
## log_GDP           0.272127   0.090148   3.019  0.00382 **
## factor(Year)2006 -0.094089   0.512703  -0.184  0.85506
## factor(Year)2008 -0.327396   0.525656  -0.623  0.53592
## factor(Year)2009 -1.064825   0.522432  -2.038  0.04626 *
## factor(Year)2011  0.104357   0.371523   0.281  0.77983
## factor(Year)2012 -0.300291   0.375966  -0.799  0.42783
## factor(Year)2013  0.047017   0.345373   0.136  0.89220
## factor(Year)2014 -0.202954   0.330004  -0.615  0.54104
## factor(Year)2015  0.194192   0.335866   0.578  0.56546
## factor(Year)2016  0.654335   0.375522   1.742  0.08692 .
## factor(Year)2017 -0.238308   0.342796  -0.695  0.48981
## factor(Year)2018  0.246002   0.304130   0.809  0.42201
## factor(Year)2019  0.371635   0.292859   1.269  0.20969
```

```
## factor(Year)2020  0.662935   0.384845   1.723  0.09048 .
## factor(Year)2021  0.340690   0.294921   1.155  0.25292
## factor(Year)2022  0.160846   0.308502   0.521  0.60416
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4433 on 56 degrees of freedom
## Multiple R-squared:  0.4244, Adjusted R-squared:  0.2496
## F-statistic: 2.429 on 17 and 56 DF,  p-value: 0.006652
```

From the above table, we can deduce that:

Higher inequality is associated with higher sustainability ratings

Coefficient on Gini = 0.0206, p = 0.0288 This means that a one-point increase in the Gini index is associated with a 0.02-point increase in the sustainability rating, after controlling for GDP and year. Although the effect size is small, it is statistically significant at the 5% level.

Wealthier countries have significantly stronger sustainability ratings

Coefficient on log_GDP = 0.272, p = 0.0038 This means that a 1-unit increase in log(GDP), roughly a 2.7× increase in actual GDP per capita, is associated with a 0.27-point increase in sustainability rating. This is a moderately large substantive effect, and the p-value indicates strong significance.

Year effects are mostly small and not significant

Most year dummy coefficients lie between −0.3 and +0.3 and are far from statistical significance ($p > 0.20$), except for 2009: ***Year 2009 coefficient = −1.06, p = 0.046*** This reflects that 2009 is an unusually low outlier year which is likely due to a smaller set of countries reporting that year. Overall, year adjustment does not meaningfully change the Gini or GDP estimates, which indicates robustness.

Model fit is modest but meaningful

R^2 = 0.424, Adjusted R^2 = 0.249 This means that The model explains 42% of variation in sustainability ratings, but only 25% after adjusting for the number of predictors, though this is expected for cross-country institutional scores, which are noisy and influenced by many unmeasured factors.

To conclude, after controlling for GDP and year, income inequality shows a small but statistically significant positive association with sustainability ratings. This suggests that more unequal countries in the dataset tend to report slightly higher sustainability performance. While this pattern does not necessarily imply causation, it indicates that inequality does not depress institutional environmental ratings in the way one might expect.

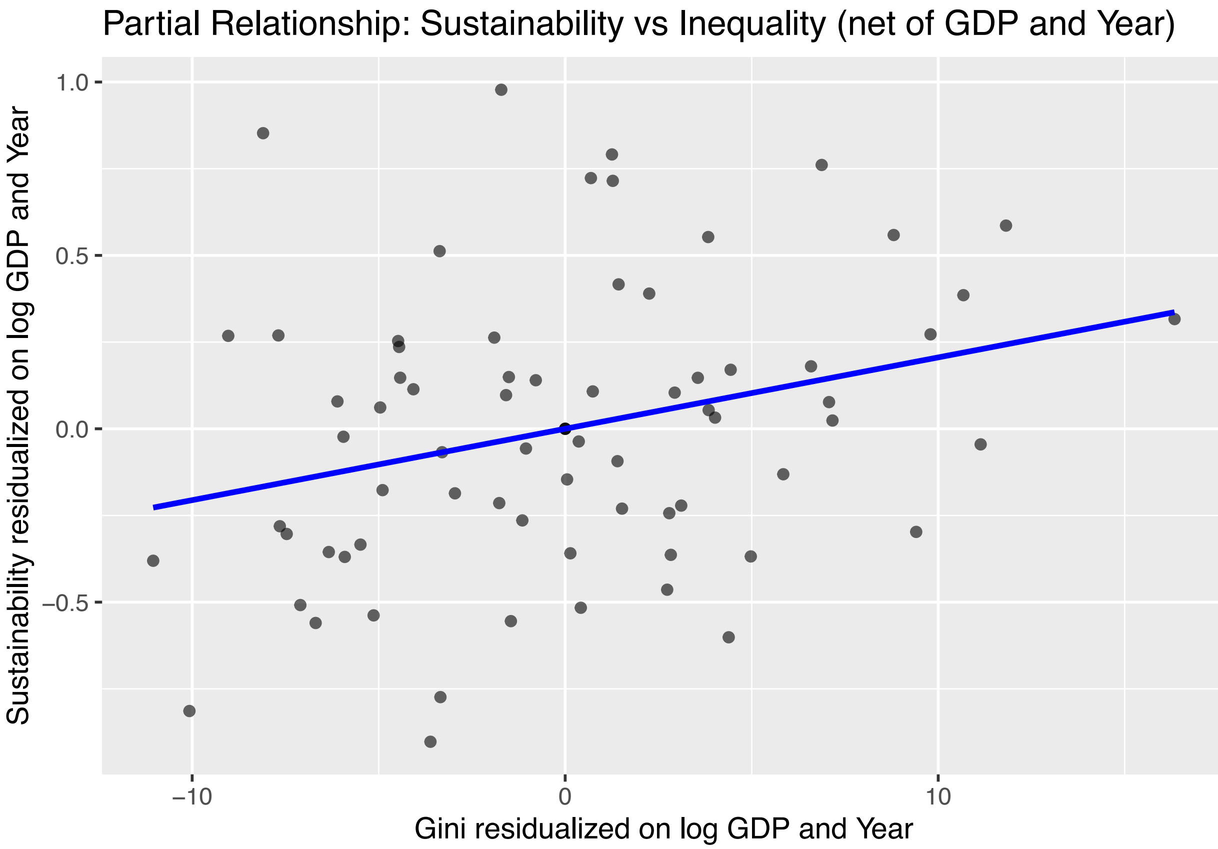
Economic development, however, displays a stronger and more consistent relationship with sustainability: countries with higher GDP per capita have substantially higher CPIA sustainability ratings. This aligns with broader empirical evidence linking economic capacity to institutional strength.

Year effects are generally not significant, suggesting that differences across reporting years do not drive the observed relationships between inequality, GDP, and sustainability.

Partial Residual Plot

To visualize the association between inequality and sustainability net of confounding factors, we constructed a partial-residual plot. Both Sustainability and Gini were residualized on log(GDP) and reporting year, and the resulting adjusted variables were plotted against each other.

```
## `geom_smooth()` using formula = 'y ~ x'
```



Key findings from the plot:

- The scatter is wide and fairly diffuse, indicating substantial variation in sustainability ratings that is not explained by inequality alone.
- Despite the noise, the fitted regression line slopes upward, echoing the positive and statistically significant coefficient from the regression model.
- Countries with higher inequality tend to have slightly higher residualized sustainability ratings even after controlling for GDP and year.
- The relationship is weak in magnitude but consistent in direction with the estimated coefficient on Gini from the main model.

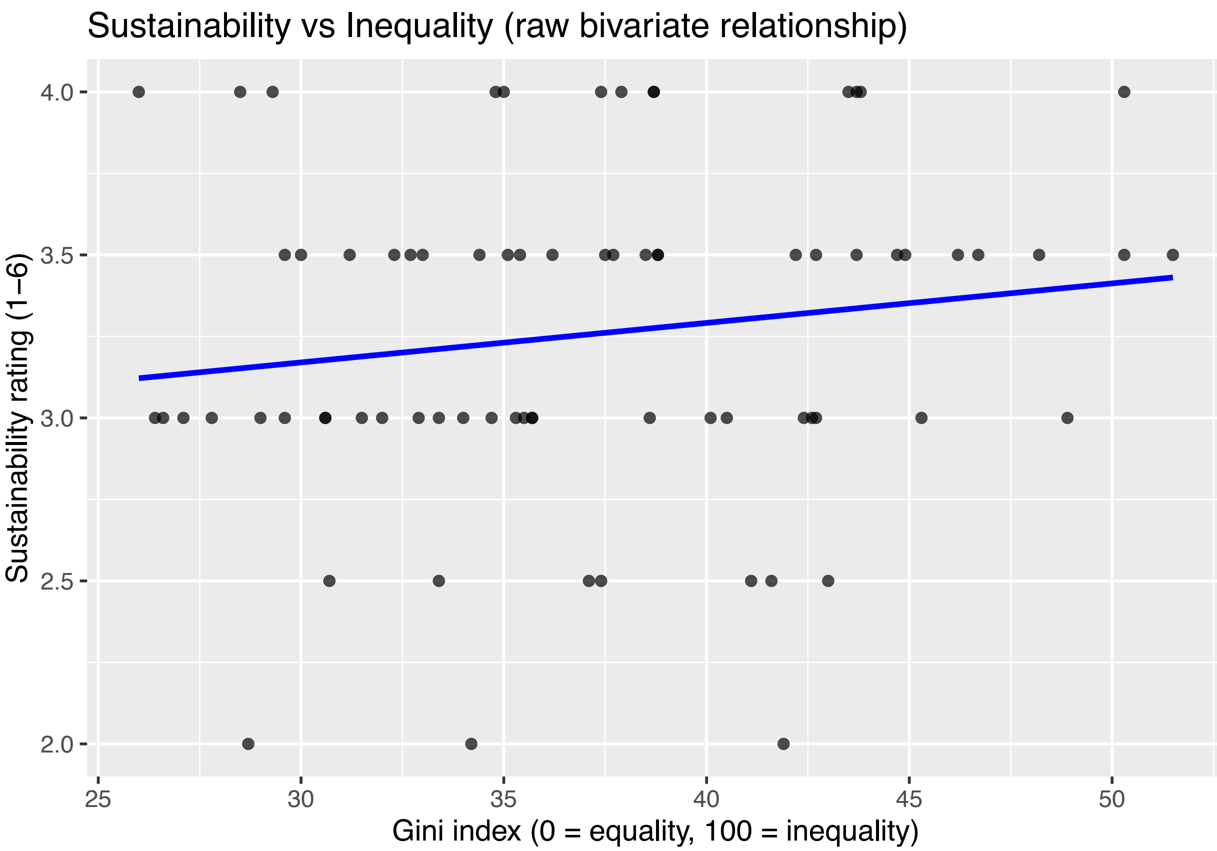
These mean that The partial-residual plot reinforces the regression finding that income inequality is positively associated with sustainability ratings, conditional on GDP and reporting year. While the association is small and the scatter shows considerable heterogeneity across countries, the upward slope indicates that inequality does not depress sustainability scores in this dataset.

However, these results do not imply causation; instead, they show that when economic development and temporal differences are accounted for, countries with higher Gini indices tend to report slightly higher environmental policy and institutional performance.

Sustainability vs Gini

```r
ggplot(rq2_panel, aes(x = Gini, y = Sustainability)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(
    x = "Gini index (0 = equality, 100 = inequality)",
    y = "Sustainability rating (1-6)",
    title = "Sustainability vs Inequality (raw bivariate relationship)"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Key findings from the plot:

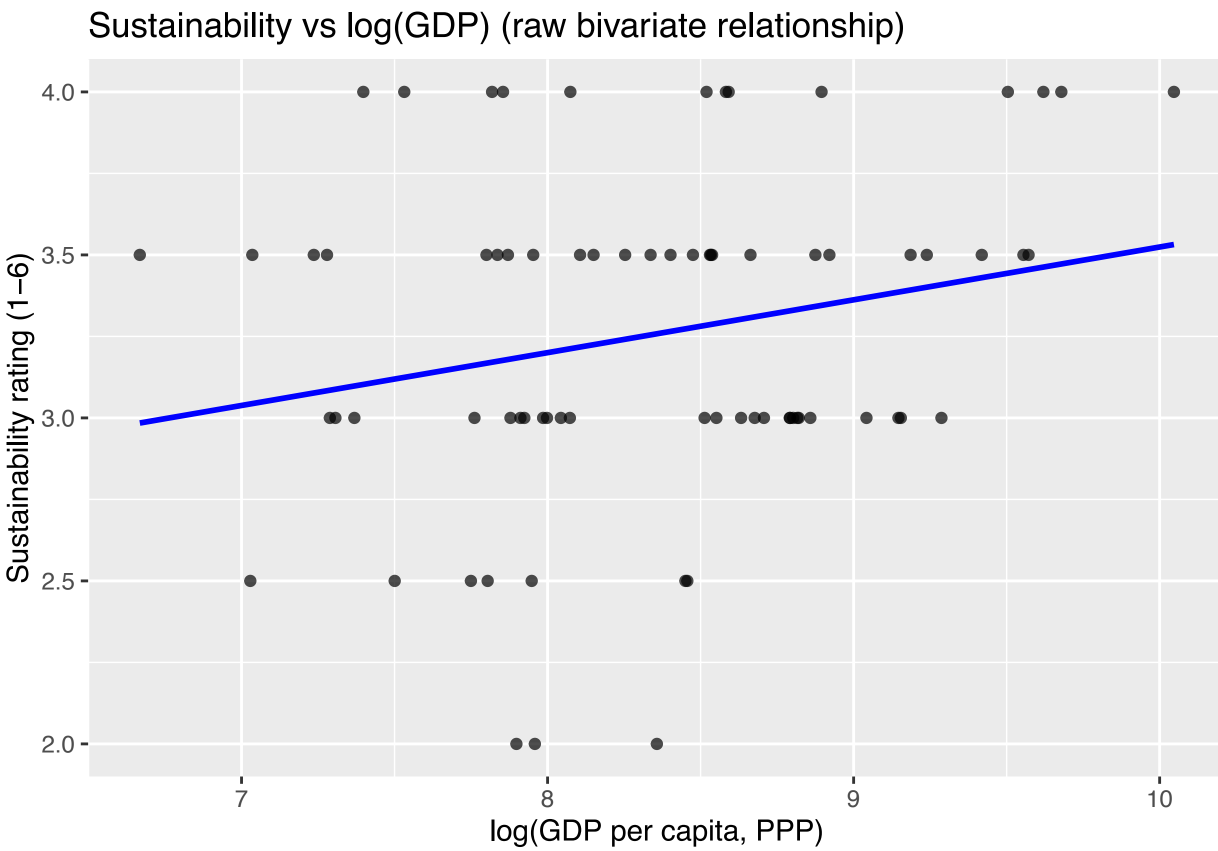
- The upward slope indicates that higher inequality is associated with slightly higher sustainability ratings in the raw data.
- The overall spread of points is wide and noisy, suggesting that inequality alone explains very little of the variation in sustainability performance.
- A large number of countries cluster at sustainability levels 3 or 3.5 regardless of inequality, indicating institutional ratings are fairly coarse.

The bivariate association mirrors the sign of the regression coefficient: more unequal countries tend to report marginally higher sustainability scores. However, the weakness of the slope and the scatter in the plot show that this relationship is not strong when examined without controls. This justifies the use of the multivariate model to account for economic development and reporting year.

Sustainability vs log(GDP)

```r
ggplot(rq2_panel, aes(x = log_GDP, y = Sustainability)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(
    x = "log(GDP per capita, PPP)",
    y = "Sustainability rating (1-6)",
    title = "Sustainability vs log(GDP) (raw bivariate relationship)"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Key findings from the plot:

- The trend line shows a stronger and more consistent positive slope than the Gini plot.
- Countries with higher log(GDP) tend to cluster at sustainability ratings of 3.5–4, while lower-income countries typically score 3 or below.
- The relationship appears more stable and less noisy than the inequality plot.

This bivariate pattern aligns closely with the regression result: wealthier countries tend to have stronger environmental policy and institutional performance. Economic capacity appears to play a more substantial role in shaping sustainability ratings than inequality does when each predictor is examined on its own.

Single year regression

Similar to RQ1, we wanted to fit a single year regression model, expecting to see improvement in data prediction. For RQ2, this corresponds to 2019, with 12 countries reporting sustainability ratings.

```
year_counts <- rq2_panel %>%
  count(Year, name = "n_countries") %>%
  arrange(desc(n_countries))

year_counts
```

```
## # A tibble: 16 x 2
##     Year n_countries
##    <dbl>       <int>
##  1  2019          12
##  2  2021          11
##  3  2018           8
##  4  2022           7
##  5  2014           5
##  6  2015           5
##  7  2013           4
##  8  2017           4
##  9  2005           3
## 10  2011           3
## 11  2012           3
## 12  2016           3
## 13  2020           3
## 14  2006           1
## 15  2008           1
## 16  2009           1
```

The model estimated is:

$Sustainability_{i,2019} = _0 + _1 Gini_i + _2log(GDP_i)+ _i. $

```
# Filter to the chosen year (2019)
rq2_2019 <- rq2_panel %>%
  filter(Year == 2019)

# Fit the cross-sectional regression
model_rq2_cross <- lm(
  Sustainability ~ Gini + log_GDP,
  data = rq2_2019
)

summary(model_rq2_cross)
```

```
##
## Call:
## lm(formula = Sustainability ~ Gini + log_GDP, data = rq2_2019)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.51785 -0.23134  0.04911  0.25453  0.47410
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  1.01484    1.51845   0.668    0.521
## Gini         0.01734    0.01435   1.208    0.258
## log_GDP      0.21678    0.13876   1.562    0.153
##
## Residual standard error: 0.3582 on 9 degrees of freedom
## Multiple R-squared:  0.2302, Adjusted R-squared:  0.05912
## F-statistic: 1.346 on 2 and 9 DF,  p-value: 0.3081
```

From the table we can see that:

- Neither Gini nor log(GDP) is statistically significant in this single-year sample.
  - Gini: estimate = 0.017, p = 0.258
  - log(GDP): estimate = 0.217, p = 0.153
- Both coefficients remain positive, matching their signs in Model 1.
- The effect sizes are similar to the panel model, but standard errors are much larger due to the smaller sample.
- The overall fit is low:
  - $R^2 = 0.23$,
  - Adjusted R^2 0.06, meaning the model explains little variation in sustainability ratings for this limited cross section.

The 2019-only regression suggests that while the direction of the relationships is consistent, showcasing by the higher inequality and higher GDP associated with slightly higher sustainability scores, the estimates are too imprecise to draw strong conclusions from a single year. The limited sample of 12 countries reduces statistical power, causing coefficients that were significant in the panel model to become non-significant here.

Therefore, Model 2 does not overturn Model 1. Instead, it shows that with only one year of data, the relationships become statistically weak even though the qualitative patterns remain the same. This reinforces the importance of using the full panel structure for institutional indicators like sustainability.

Conclusion

Across countries, sustainability ratings tend to be higher in wealthier nations and slightly higher in more unequal ones. The multivariate panel model shows that both income inequality and economic development are positively associated with environmental sustainability ratings after accounting for year differences. The effect of inequality is statistically significant but modest in size, whereas GDP demonstrates a stronger and more robust association.

The partial residual analysis indicates that, net of GDP and year, the positive relationship between inequality and sustainability persists, though with considerable variability across countries. Simple bivariate plots confirm that GDP is a more substantial predictor of sustainability than inequality when each is examined independently.

The single-year model for 2019, although directionally consistent with the panel results, suffers from limited sample size and produces imprecise estimates. As a result, the evidence supporting the role of inequality is clearer in the full dataset than in a single-year cross-section. Overall, economic development appears to be the key driver of sustainability ratings, while inequality shows a small but positive association that warrants cautious interpretation.